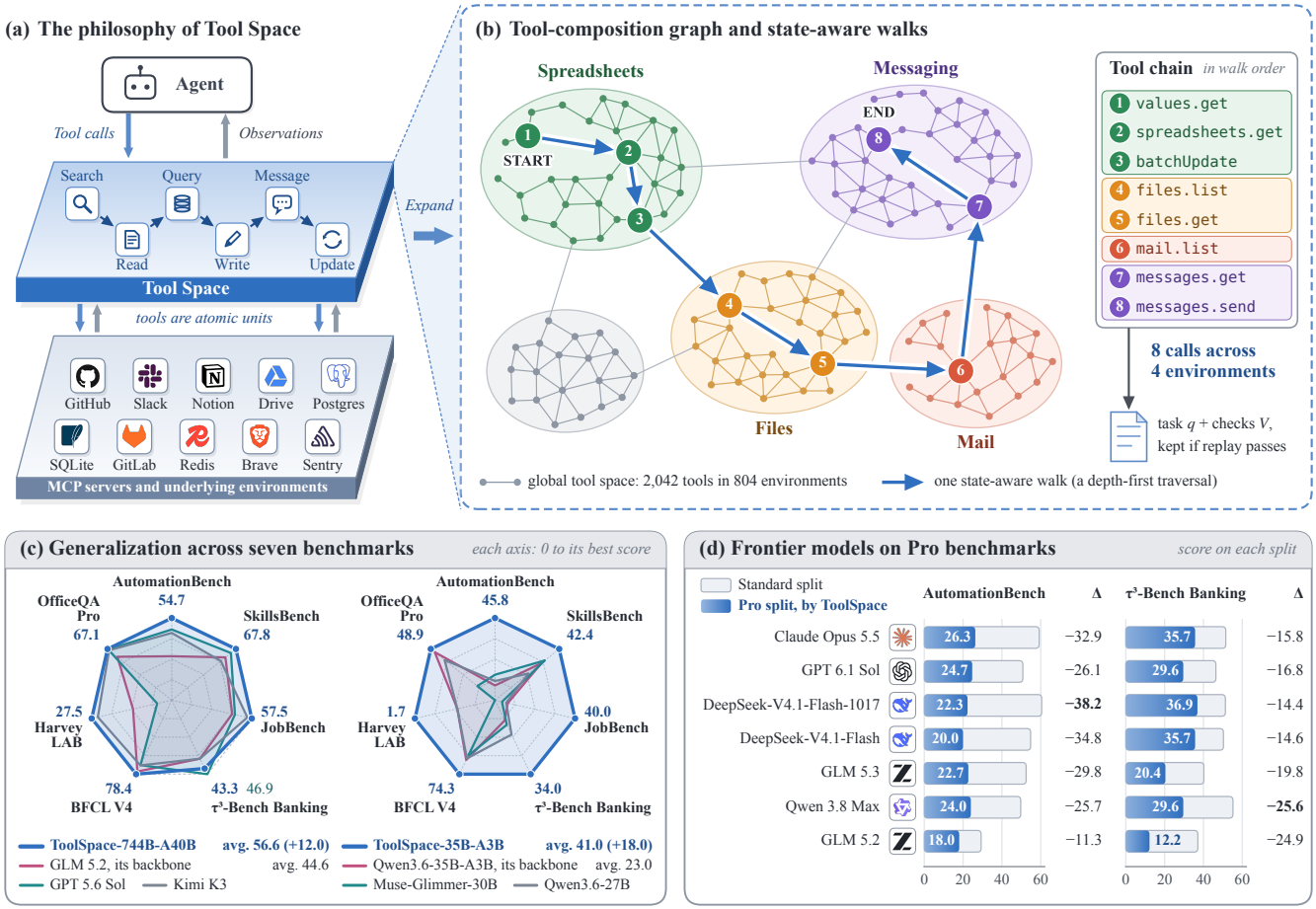

(a) The philosophy of Tool Space
Agent
Tool calls
Observations
Search
Query
Message
Read
Write
Update
Tool Space
tools are atomic units
GitHub
Slack
Notion
Drive
Postgres
SQLite
GitLab
Redis
Brave
Sentry
MCP servers and underlying environments
Expand
(b) Tool-composition graph and state-aware walks
Spreadsheets
Messaging
START
END
Files
Mail
Tool chain in walk order
1 values.get
2 spreadsheets.get
3 batchUpdate
4 files.list
5 files.get
6 mail.list
7 messages.get
8 messages.send
8 calls across 4 environments
task q + checks V, kept if replay passes
global tool space: 2,042 tools in 804 environments
one state-aware walk (a depth-first traversal)
(c) Generalization across seven benchmarks
each axis: 0 to its best score
AutomationBench
OfficeQA Pro
SkillsBench
JobBench
τ³-Bench Banking
BFCL V4
Harvey LAB
54.7
67.8
57.5
43.3
46.9
78.4
27.5
67.1
45.8
42.4
40.0
34.0
74.3
1.7
48.9
ToolSpace-744B-A40B avg. 56.6 (+12.0)
GLM 5.2, its backbone avg. 44.6
GPT 5.6 Sol
Kimi K3
ToolSpace-35B-A3B avg. 41.0 (+18.0)
Qwen3.6-35B-A3B, its backbone avg. 23.0
Muse-Glimmer-30B
Qwen3.6-27B
(d) Frontier models on Pro benchmarks
score on each split
Standard split
Pro split, by ToolSpace
AutomationBench
Δ
τ³-Bench Banking
Claude Opus 5.5
GPT 6.1 Sol
DeepSeek-V4.1-Flash-1017
DeepSeek-V4.1-Flash
GLM 5.3
Qwen 3.8 Max
GLM 5.2
26.3
24.7
22.3
20.0
22.7
24.0
18.0
−32.9
−26.1
−38.2
−34.8
−29.8
−25.7
−11.3
35.7
29.6
36.9
35.7
20.4
29.6
12.2
−15.8
−16.8
−14.4
−14.6
−19.8
−25.6
−24.9
0
20
40
60